

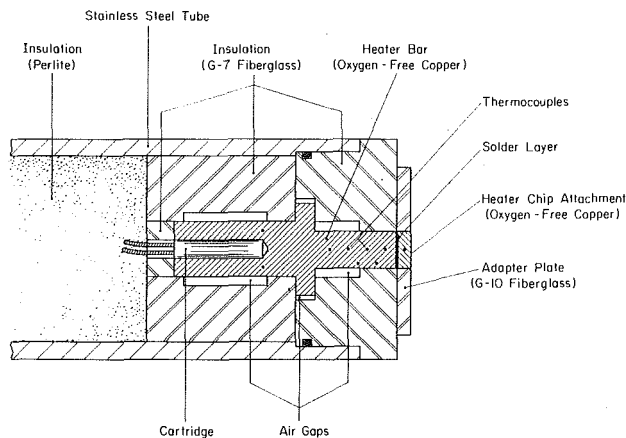


Fig. 2 Sectional diagram of the test heater

The light source provided the required luminosity during short flash durations, resulting in a 1/3000 second exposure time.

3 Numerical Results

Practical Design Considerations. The design of finned surfaces for electronic chips must satisfy several stringent requirements. Of paramount importance are: (1) increasing critical heat flux above the maximum expected chip heat flux, (2) maintaining chip temperature below an acceptable limit, (3) preventing the temperature excursion at incipient boiling commonly encountered with low contact angle fluids, and (4) limiting the accumulation of vapor on the chip surface in order to minimize the growth of a thick vapor bubble layer on multichip modules. Although extended surfaces can be expected to satisfy the requirement of increasing heat dissipation from the chip, the choice of a fin geometry may promote many adverse effects, making requirements (2), (3), and (4) hard to achieve. Thus, increasing heat dissipation by finning must be accompanied by such important design changes as subcooling the dielectric fluid or using a fluid with a boiling point much lower than the maximum acceptable chip temperature.

Preventing the incipience excursion leaves very few acceptable geometries for fin design. Anderson and Mudawar (1989) examined the feasibility of enhancing chip cooling performance by machining microstructures (rectangular microgrooves or

square microstuds) into the surface. They reported a significant reduction in wall superheat and CHF enhancement by factors as high as 2.51 over a smooth surface; yet, these attractive results were coupled with incipience temperature excursions as high as 14.3°C. Mudawar and Anderson (1990) attempted to reduce the incipience excursion on a vertically mounted chip by the high thermal mass of a tapered (pyramid-shaped) fin with microgrooves machined into its perimeter. Boiling was observed to commence near the base. The inclined downward facing side of the tapered fin allowed bubbles from the base to rapidly activate boiling on the entire surface, resulting in a noticeable excursion in the base temperature. Thus, they suggested that enhanced electronic cooling surfaces be designed to spread nucleation in a gradual manner with increasing heat flux. They concluded that chip extended surface geometries should be limited to horizontal, cylindrical shapes. The present study examines the cylindrical shape as a basic building block in the design of enhanced surfaces.

Numerical Results for Fin Effectiveness. Numerical heat conducting codes were used to predict the effectiveness of smooth horizontal cylindrical fins in saturated FC-72 and FC-87 having boiling points of 56°C and 30°C at one atmosphere, respectively. Fin effectiveness solutions were generated for a cylindrical fin in FC-72 enhanced circumferentially by either microgrooves or microstuds. Various fin materials were examined to evaluate the influence of thermal conductivity on fin effectiveness. Copper, aluminum and brass, having conductivities of 391, 240 and 110 W/m·K, respectively, were the reference materials for the numerical study.

The simulation of fin heat transfer during natural convection and nucleate boiling was simplified by assuming constant thermal conductivity, fluid temperature and base temperature, and by ignoring radiative heat transfer. The cylindrical fins were assumed to experience axisymmetric heat transfer according to the equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0 \quad (1)$$

where x is the distance from the base measured along to the fin axis and r emanates radially from the axis.

The calculation domain was divided into a uniform grid of control volumes following the procedure suggested by Patankar (1980). Following Haley and Westwater (1966) and Nakayama et al. (1984), the local boiling heat transfer flux on the fin surface at a distance x from the base was assumed equal

Nomenclature (cont.)

Ra_l = Rayleigh number for a vertical plate of height $l = \frac{g\beta_f(\Delta T)l^3}{\nu_f\alpha_f}$
 ΔT = $T_w - T_\infty$
 ΔT_h = wall temperature drop at incipience
 ΔT_i = $T_{w,i} - T_\infty$ (or $T_{w,i} - T_{sat}$ for saturated boiling)
 ΔT_m = wall superheat at CHF
 ΔT_{sat} = wall superheat = $T_w - T_{sat}$
 ΔT_{sub} = bulk liquid subcooling = $T_{sat} - T_\infty$
 T_w = mean temperature at the base of enhanced surface attachment
 $T_{w,i}$ = incipient boiling temperature at the base of enhanced surface
 T_∞ = bulk liquid temperature

T_{85} = 85°C base temperature
 w = characteristic fin width and fin clearance of microgrooves and microstuds = 0.305 mm
 x = distance from the base of the cylindrical fin measured along axis
 α_f = thermal diffusivity of liquid
 β_f = expansion coefficient of liquid
 γ = bubble departure diameter scaling factor
 ϵ_e = effectiveness of finned element = $[Q_f + q_b(A_e - A_b)] / [q_b A_e]$
 ϵ_f = effectiveness of cylindrical fin = $\frac{Q_f}{q_b A_b}$
 ϵ_{chip} = overall effectiveness of enhanced surface

η_e = efficiency of finned element = $[Q_f + q_b(A_e - A_b)] / [q_b(A_f + A_e - A_b)]$
 η_f = efficiency of cylindrical fin = $\frac{Q_f}{q_b A_f}$
 λ_D = Taylor "most dangerous wavelength" = $2\pi \left[\frac{3\sigma}{(\rho_f - \rho_g)g} \right]^{1/2}$
 λ_T = Taylor wavelength = $2\pi \left[\frac{\sigma}{(\rho_f - \rho_g)g} \right]^{1/2}$
 ν_f = kinematic viscosity of liquid
 ρ_f = density of liquid
 ρ_g = density of vapor
 σ = surface tension